

Linear prediction analysis of speech based on pole-zero representation

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THURSDAY, 6 NOVEMBER 1975

GOLDEN GATE ROOM, 2:00 P.M.

Session VV. Speech Communication VIII: Speech Analysis 2

John D. Markel, Chairman

*Speech Communication Laboratory, Santa Barbara, California 93109**Contributed Papers*

2:00

VV1. Production and utilization of junctural acoustic cues under different speaking conditions. M. S. Harris (Bell Laboratories, Murray Hill, NJ 07974)

At a previous meeting we reported perceptual results concerning so-called internal open juncture phrases in three situations: spoken in isolation, embedded in proper sentences, and embedded in nonsense sentence frames in which the other member of the minimal pair made sense. The rate of correct responses decreased in the above order. In nonsense sentences, in particular, the rate of correct responses was quite low. Acoustic analysis of the production of these minimal pair phrases in the different environments is currently under way. Initial inspection of the duration data indicates features in the isolated phrases that are more exaggerated than those same features when the phrases are in proper sentences. It appears, because the phrases in inappropriate sentence frames were also emphasized, that the speaker took extra care in articulating these phrases. This is particularly interesting because the listeners did not detect or choose to use the additional acoustic cues provided by the feature emphasis.

2:12

VV2. Some acoustic characteristics of syllable nuclei in conversational speech. M. A. Earle and L. L. Pfeifer (Speech Communications Research Laboratory, Inc., 800-A Miramonte Drive, Santa Barbara, CA 93109)

Approximately 5 min of conversational speech from an interview of one male speaker of Midwestern American was transcribed and digitized. The digitized speech is bandlimited at 5 kHz, sampled at 10 kHz, and quantized to 12 bits per sample. Steady-state portions of the vowels and initial and final portions of the diphthongs are located by visual inspection of a time-synchronized display of the waveform, spectral peaks, and the rms of the signal. Formant parameters derived from the inverse filter method of linear prediction are then extracted from four consecutive spectral samples representing 29.6 msec of the steady state of each vowel or of the most stable region in the initial and final portions of each diphthong. The effect of various segmental environments on some acoustic parameters, such as formant frequencies, is discussed. A comparison is made of the vowel and diphthong samples in terms of formant frequency means and standard deviations. [This research is supported by the Air Force Office of Scientific Research under Contract F44620-74-C-0034.]

2:24

VV3. Dental flaps and vowel duration in English. Robert A. Fox and Dale Terbeek (Department of Linguistics, University of Chicago, Chicago IL 60637)

This experiment investigated dental flaps and their effect upon preceding vowel duration in English. Twenty speakers read a randomized list of 40 disyllabic words, 20 with medial /d/. Most tokens of /t/ and /d/ were pronounced as flaps, as expected. The data, collected from spectrograms, included 1st and 2nd vowel duration, presence or absence of flap (as opposed to stop [t] or [d]), duration of consonant, and extent of

voicing during the flap. Our results provide evidence bearing on two phonological questions: (1) Does the rule which changes /t/ and /d/ to flaps (F) precede or follow the rule which lengthens vowels before voiced consonants (VL)? If VL precedes F, the vowels preceding flaps with underlying /d/ will have greater duration than flaps with underlying /t/. If F precedes VL, there will be no such difference. (2) Do underlying /t/ and /d/, in fact, give rise to the same voiced flap consonant, as is usually assumed? The data suggests that not all /t/ flaps and /d/ flaps are identical, nor are they necessarily voiced.

2:36

VV4. Linear prediction analysis of speech based on a pole-zero representation. B. S. Atal (Bell Laboratories, Murray Hill, NJ 07974) and M. R. Schroeder (Drittes Physikalisches Institut, University of Goettingen, West Germany)

Speech analysis and synthesis by linear prediction is based on the assumption that the short-time spectral envelope of speech can be represented by a number of poles. An all-pole representation does not provide an accurate description of speech spectra, particularly for nasal and nasalized sounds. In this paper, a method is presented for characterizing speech in terms of the parameters of a pole-zero model. The basic approach used in the method is similar to one used for the all-pole case: the parameters of the pole-zero model are obtained by minimizing the mean-squared prediction error over the analysis interval. It is shown that, although the equations resulting from the minimization are nonlinear, both poles and zeros can be determined without using an iterative procedure. The method has been applied to real speech data and the results show that the speech spectra derived from the pole-zero model agree very closely with the actual spectra derived by direct FFT analysis. Audible distortion due to inadequate treatment of zeros in the all-pole case disappears with pole-zero representation.

2:48

VV5. Duration of English stops in prestressed position. Victor W. Zue (Department of Electrical Engineering and Computer Science and Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, MA 02139)

Durational characteristics of English stops in prestressed position, both in singleton and in clusters, are examined in detail using a data base designed for controlled acoustic phonetic research. [Victor W. Zue, "Acoustic phonetic data base for the study of selected English consonants, consonant clusters, and vowels," J. Acoust. Soc. Am. 57, S34(A) (1975)]. The study being presented here is based on the analysis of some 1200 stops spoke by 3 male speakers. Results indicate systematic variations of closure, voice-onset time, frication, and aspiration durations as a function of: (1) the voicing characteristics of the stop; (2) the place of articulation of the stop; and (3) the phonetic environment (i.e., the influence of the following vowel for singleton stops, and the influence of the preceding fricative and/or the following sonorant for clusters). Results on inter- and intraspeaker variability will also be discussed.